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PARAMETRIC EMPIRICAL BAYES METHODS FOR ECOLOGICAL APPLICATIONS[1]

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Abstract. This paper reviews parametric empirical Bayes (PEB) methods and their application to ecological data. One of the main objections to Bayesian statistics is the "subjectivity" caused by specifying parameters. Attempting to remove that subjectivity, the PEB philosophy uses the Bayesian mathematical framework, but then estimates, rather than specifies, all unknown parameters. Three examples are given to illustrate PEB methods. The first is a PEB analysis for monitoring the yearly population of harbor seals, the second uses simulated data and shows how a PEB analysis combines single capture–recapture estimates, and the third is a PEB method for spatial patch analysis of data from a transect through grassland vegetation. From the examples, the general methodology and philosophy of a PEB analysis are explained.

Key words: Bayesian inference; combining capture–recapture experiments; ecological statistics; edge-detection; hierarchical models; image analysis; population monitoring.

INTRODUCTION

The term "empirical Bayes" was coined by Robbins (1955). In the broadest sense, the underlying goal of an empirical Bayes method is to use Bayesian methods without fully specifying the prior, either by estimating the prior or its parameters. There are essentially two approaches, nonparametric and parametric. For the nonparametric version, see Maritz and Lwin (1989). The nonparametric approach requires very large sample sizes. When sample sizes are smaller, a parametric approach is indicated. This paper introduces parametric empirical Bayes (PEB) methods, with seminal papers by Efron and Morris (1973, 1975), and other introductions including Efron and Morris (1977) and Casella (1985, 1992). The fullest exposition is given by Morris (1983*a*), with interesting discussion following his paper.

Background

Ecologists are often required to make many estimates from similar data. This situation can be depicted by Table 1. One example is a program to monitor the abundance θ_i of a population in year i, with yearly estimates of abundance $\hat{\theta}_i$ from data $y_{i1}, y_{i2} \ldots y_{in_i}$, where there are n_i replicate samples in year i. Suppose that simple random sampling was used. Then sampling theory (Cochran 1977, Thompson 1992) indicates that $\hat{\theta}_i$ is the sample mean and that $\hat{\theta}_i$ is unbiased for θ_i, with an associated estimate of variance. This is done year after year, up to year k. As another example, consider estimating a regression slope parameter for several geographic populations, where we might, say, regress vegetation biomass on soil nitrogen at each location. Then there are several estimates of the slope parameter θ_i, with an associated measure of precision for each population i. For both of these examples, the question arises, can we get better estimates of θ_i by borrowing information from data in those populations other than the ith one? For the Bayesian and empirical Bayesian, the answer is yes.

I will first describe three applications to give an intuitive feel for the PEB approach. Then, I will provide some of the philosophy and statistical mechanics used in the applications.

ECOLOGICAL APPLICATIONS

Monitoring harbor seals

Since the Exxon Valdez oil spill of 1989, the Alaska Department of Fish and Game has been monitoring harbor seals (*Phoca vitulina richardsii*) at 25 haul-out sites in the central Prince William Sound. The goal is to estimate the average number of seals hauled out during low tide during the molting period of late summer for each year from 1989 through 1994. This will be an unknown proportion of the true number of seals, but it is assumed to be a constant proportion from year to year. Seven to 10 replicate flights to count seals were conducted each year.

Not all sites could be observed on each flight, due to bad weather, etc., so bootstrapping was used to estimate the average number of seals that were hauled out at the 25 locations. For each bootstrap iteration, the mean for each site was calculated, and then summed for all 25 sites. Hence, we have a situation as given in Table 1. In the notation of Table 1, $\hat{\theta}_i$ is the mean of the 2000 bootstrap iterations for each year, and the

[1] Manuscript received 14 August 1995; revised 14 November 1995; accepted 21 November 1995; final version received 16 January 1996.

[2] For reprints of this group of papers on Bayesian inference, see footnote 1, page 1034.

TABLE 1. A situation often faced by ecologists is having similar data and making several to many estimates from those data. The data are presented as univariate y (although they could be multivariate), with the possibility of additional covariates (the x's). The goal is to make an inference on some parameter θ_i for each population i. The data are used to make an estimate $\hat{\theta}_i$ with an associated variance σ_i^2.

Data	Parameter	Estimate	Variance
$y_{11}, y_{12}, \ldots, y_{1n_1}$ $x_{11}, x_{12}, \ldots, x_{1n_1}$	θ_1	$\hat{\theta}_1$	σ_1^2
$y_{21}, y_{22}, \ldots, y_{2n_2}$ $x_{21}, x_{22}, \ldots, x_{1n_2}$	θ_2	$\hat{\theta}_2$	σ_2^2
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$y_{k1}, y_{k2}, \ldots, y_{kn_k}$ $x_{k1}, x_{k2}, \ldots, x_{kn_k}$	θ_k	$\hat{\theta}_k$	σ_k^2

standard error was estimated by the standard deviation of the bootstrap iterations. The means and standard errors for each year are given in Table 2.

From Table 2, there appears to be a decreasing trend in numbers over the years, and one is tempted to put the $\hat{\theta}_i$ into a linear regression. In fact, that is how we shall model the data. There are two sources of variability. One is due to the variation of replicate counts within year (I will call this the sampling variance). The other is due to variation around some trend line among years (I will call this regression variance). This model has been discussed in the context of breeding bird surveys by Link et al. (1994). There are now two reasonable estimates for θ_i. One is the yearly sample mean, and the other is from the regression line of the yearly sample means, call it $\hat{\mu}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, where x_i is the ith year and $\hat{\beta}_0$ and $\hat{\beta}_1$ are the regression parameters fit to the yearly means. Which one is to be preferred, $\hat{\theta}_i$ or $\hat{\mu}_i$? Really, $\hat{\mu}_i$ is estimating the mean value of a linear trend, but the number that are actually there (the true value θ_i) will not fall exactly on the line, so it seems that $\hat{\theta}_i$ might be preferred. Yet, suppose that the regression variance (denoted δ^2) of the θ_i around the regression line is much less than the sampling variance σ_i^2; then $\hat{\mu}_i$ might be preferred. But how much smaller should δ^2 be than σ_i^2? In fact, there is no reason to choose between the two, but instead combine them,

$$\tilde{\theta}_i = (1 - B_i)\hat{\theta}_i + B_i\hat{\mu}_i, \qquad (1)$$

where $0 \leq B_i \leq 1$ is a "shrinking" coefficient that ensures that $\tilde{\theta}_i$ is somewhere between $\hat{\theta}_i$ and $\hat{\mu}_i$.

Eq. 1 shows up naturally in a PEB framework; the mathematical details are given in the Appendix and by Morris (1983*a*), who also details how B_i, μ_i, δ^2, and var($\tilde{\theta}_i$) are estimated. For the harbor seal data, $\hat{\mu}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ with $\hat{\beta}_0 = 837.6$, $\hat{\beta}_1 = -20.0$, $\sqrt{\delta^2} = 56.8$, and year x_i was re-indexed to start at 0. There are several things to notice about Eq. 1. If σ_i^2 and δ^2 were known, then $B_i = \sigma_i^2/(\sigma_i^2 + \delta^2)$, so how far the PEB estimate shrinks toward the regression line depends on

TABLE 2. Estimation and variances for harbor seal data. The estimated trend is $\hat{\mu}_i$. The bootstrap estimate is $\hat{\theta}_i$ and the PEB estimate is $\tilde{\theta}_i$. The estimated shrinkage factor is B_i.

Year	$\hat{\mu}_i$	$\hat{\theta}_i$	$\tilde{\theta}_i$	SE($\hat{\theta}_i$)†	SE($\tilde{\theta}_i$)†	B_i
1989	838	815	816	21.6	21.4	0.06
1990	818	779	784	33.5	32.4	0.13
1991	798	920	897	43.7	45.6	0.19
1992	778	760	762	27.7	26.7	0.10
1993	758	774	771	38.5	36.6	0.16
1994	738	679	696	64.6	60.3	0.28

† SE = standard error.

the sampling variance relative to the regression variance. This is extremely appealing; as the sampling variance increases, the PEB estimate shrinks toward the regression line, or, if the regression variance is much larger than the sampling variance, the PEB estimate is virtually the same as the yearly sample estimate. For example, see the estimates of B_i in relation to standard errors of $\hat{\theta}_i$ in Table 2. The PEB method "borrows strength from the ensemble" (Morris 1983*a*), by shrinking toward the regression. Fig. 1 shows this graphically. Notice that especially in years 1991 and 1994, with larger sampling variances, there is more shrinkage. Another interesting point is that the PEB standard errors are generally smaller than the yearly sampling standard errors, so that credibility intervals (Bayesian word for confidence interval) are generally shorter than the sampling confidence interval.

While some ecologists and statisticians may dislike Bayesian data analysis, almost everyone agrees that

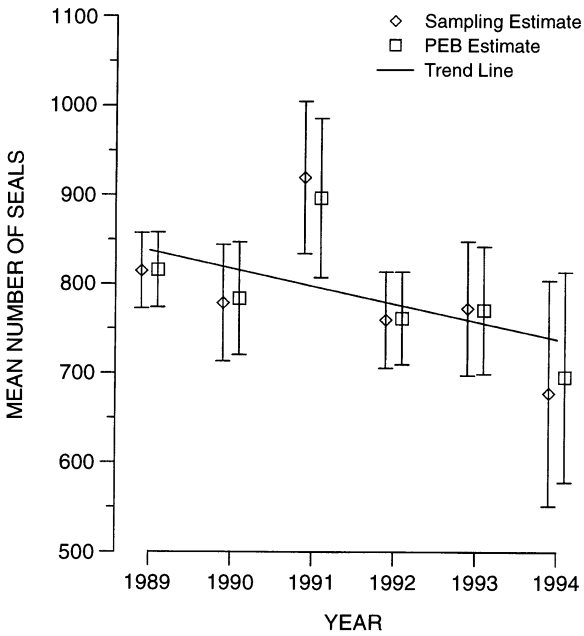

FIG. 1. Parametric empirical Bayes estimates compared to single yearly sampling estimates for harbor seal data. The vertical bars are the 95% confidence intervals (for the sampling estimates) and 95% credibility intervals (for the PEB estimates).

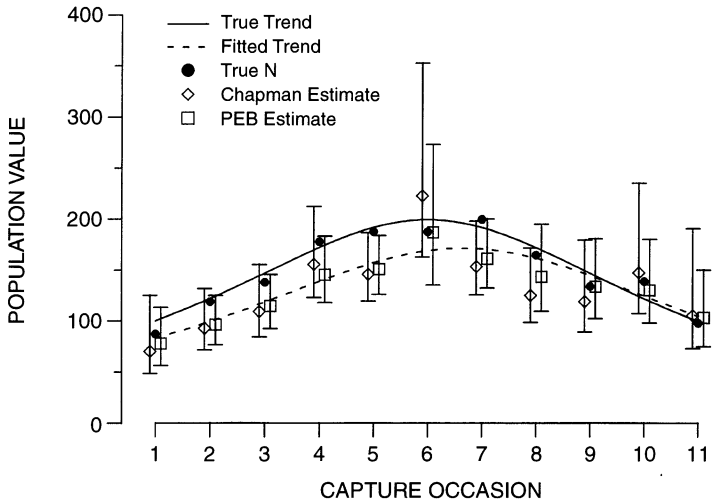

FIG. 2. Parametric empirical Bayes estimates compared to single Chapman estimates for simulated capture–recapture data. The vertical bars are the 95% confidence intervals (for the sampling estimates) and 95% credibility intervals (for the PEB estimates).

Bayesian credibility intervals are much easier to explain and use than confidence intervals. This is because the posterior distribution contains all information about the unknown parameter, so confidence statements can be made directly from the posterior distribution. Also, if interest centers on some function of the parameter, it is generally easy to simulate the credibility interval. As an example, suppose that, besides estimating each θ_i, we also want to estimate the maximum yearly change in harbor seal abundance over the 6-yr period. This can be written as $\tau = \max(|\theta_i - \theta_{i+1}|)$ for $i = 1, \ldots, 5$. Estimates and confidence intervals would be difficult using classical sampling theory, but bootstrapping could be used. The PEB method uses the posterior distribution. Simply simulate realizations from a normal distribution for each year with the mean given by the PEB estimate $\tilde{\theta}_i$ and variance given by $\mathrm{var}(\tilde{\theta}_i)$ and calculate τ for each simulation. The mean of the simulations is an estimate for τ and the percentiles from the simulations give the credibility interval. Using the values from Table 2 and 2000 Monte Carlo simulations, the estimate is $\hat{\tau} = 155$ with a 95% credibility interval of 68 to 253.

TABLE 3. Simulated capture–recapture data: θ_i is the simulated population from a Poisson distribution with mean μ_i in the ith experiment. The number of marks m_i are 0.3 of the θ_i. The number of captures c_i were simulated from a $\mathrm{Bin}(\theta_i, 0.3)$, and the number of marked recaptures r_i were simulated from a $\mathrm{Hyp}(\theta_i, m_i, c_i)$.

Experiment i	c_i	r_i	m_i	θ_i	μ_i
1	20	7	26	87	100
2	38	14	35	119	122
3	41	15	41	138	147
4	60	20	53	178	172
5	66	25	56	188	192
6	58	14	56	188	200
7	65	25	60	200	192
8	47	18	49	165	172
9	40	13	40	134	147
10	45	12	41	139	122
11	31	8	29	98	100

Combining single capture–recapture estimates

The next example uses simulated data so that the true values are known. Suppose that there are to be 11 single capture–recapture experiments, and assume that they can be related in some way. For example, they might be 11 single capture–recapture experiments through time, or they might be from 11 different locations ordered along some ecological or spatial gradient. Let $x_i = i - 1$ for the $i = 1, \ldots, 11$ capture–recapture experiments. Then, suppose they are related to each other by the following: the mean number of total animals for each experiment is $\mu_i = 1/(0.01 - 0.002x_i + 0.0002x_i^2)$. This curve, the true trend, can be seen in Fig. 2. The actual number of animals will not fall exactly on this curve, however, so simulate the true θ_i from a Poisson distribution $\theta_i \sim \mathrm{Poi}(\mu_i)$. Suppose that the number of marked animals in the population m_i is some proportion π of θ_i, and suppose the number captured c_i in each experiment follows a Binomial distribution: $c_i \sim \mathrm{Bin}(\theta_i, p)$. Finally, suppose that the experiments are done without replacement so that the number of marked recaptures r_i follows a hypergeometric distribution $r_i \sim \mathrm{Hyp}(\theta_i, m_i, c_i)$. One set of data was simulated as described above with $\pi = 0.3$ and $p = 0.3$, and is given in Table 3.

A nearly unbiased estimate for each θ_i is the estimate of Chapman (1951),

$$\hat{\theta}_i = \frac{(m_i + 1)(c_i + 1)}{(r_i + 1)} - 1.$$

Again, the situation of Table 1 occurs. As in the harbor seal example, we consider two sources of variability. One is σ_i^2 due to estimating θ_i, but the θ_i are also random

with some variance, again call it δ^2, around a trending mean. A PEB shrinkage estimator was also applied. The details of the PEB estimates and credibility intervals are given in the Appendix. Notice for example, in Fig. 2, how the PEB estimates shrink toward the fitted trend line. There is a large amount of shrinkage on capture occasion 6 because the estimated variance of the Chapman (1951) estimate is large. Generally, shrinkage helps move the estimate toward the true value, as it does on all capture occasions except occasion 4. As in the harbor seal example, the PEB credibility intervals are shorter than the Chapman (1951) confidence intervals, which uses the method of Jensen (1989). On capture occasion 5, the PEB estimate moves closer to the true value, but because it has a shorter credibility interval, it does not quite cover the true value, nor does the Chapman (1951) confidence interval. However, on capture occasion 7, the PEB credibility interval does capture the true value, while the Chapman (1951) confidence interval does not.

To demonstrate that this method can increase the precision of the estimate, I simulated 91 data sets like that in Table 3 with $\pi = 0.3$ and $p = 0.3$, yielding 1001 θ_i's. For each of the 91 data sets, I computed the 11 Chapman (1951) estimates and 11 95% confidence intervals as described by Jensen (1989). I also computed the PEB estimates and credibility intervals. The average length of the confidence interval for the single capture–recapture estimate was 135.2, while the average length of the PEB credibility interval was 99.3. The actual coverage for the confidence intervals described by Jensen (1989) was 0.930, while the coverage for the PEB credibility intervals was 0.966. Thus, the PEB credibility intervals were about 73% as long as the single capture–recapture confidence intervals, yet gave better coverage! Jensen (1989) showed his confidence intervals were as short as 68% the length of those described by Chapman (1951). In terms of actual mean squared error, taking the average of $(\hat{\theta}_i - \theta_i)^2$ for all 1001 values gave 958.1 for the single capture–recapture estimates, and taking the average of $(\tilde{\theta}_i - \theta_i)^2$ gave 468.0 for the PEB estimates, showing that the PEB estimates were in general much closer to the true value than the single Chapman estimates. This also explains why the PEB method had shorter credibility intervals yet better coverage.

This method might be used as an alternative to the Jolly-Seber method (Jolly 1965, Seber 1965) for open populations when it is difficult to obtain the capture history for each animal, or when there are one or more obvious features that tie together the single capture–recapture experiments. Also, as in the harbor seal example, it is easy to get estimates and credibility intervals if one is interested in some function of the $\hat{\theta}_i$'s. The value of such procedures should be apparent when asking whether or not a population ever exceeds a critical threshold. Gazey and Staley (1986) have such an example using the classical Bayesian framework to estimate minimum population size for a closed population.

Image analysis for transect data

One of the goals of a statistical analysis of transect data is to determine "edges," which have more frequently been called "change-points" in the statistical literature (Lombard 1989). Examples of statistical analysis of these transects in the ecological and environmental literature include Webster (1973), Wierenga et al. (1987), Ludwig and Cornelius (1987), and Orlóci and Orlóci (1990). These methods generally use a moving window to locate areas of high variability as indicating the presence of a change-point. I will consider a PEB adaptation of Bayesian analysis of images (Besag 1986, Cressie 1991:509–521), from which edges, as well as other features, can be determined for the data.

The data consist of percent vertical cover in a chalk grassland in The Netherlands. The data have been described by Ver Hoef et al. (1989). Along a 30-m transect, each 50-cm segment was photographed from ground level through a strip of vegetation that was 10 cm deep. Each segment was backed by a white screen with a vertical scale on it. The 60 photographs were digitized and each was divided into five plots (each plot a vertical column 1.5 m tall by 10 cm wide), and image analysis was used to compute the percentage of nonwhite pixels, making a total of 300 data values of percent vertical cover along the transect. The data were log-transformed (with 1 added because of zeros) due to skewness of the data, and only the first 100 data points are given in Fig. 3.

The data $\{y_i : i = 1, 2, \ldots, n\}$ are modeled as follows. At each location, the datum y_i is observed, and assumed to come from a normal distribution with mean θ_i. Again, the situation of Table 1 arises. Only here, there is a parameter to be estimated for every single datum! Obviously, there is no estimate of variance in the classical sense, and only a hierarchical model allows inference. To make inference on the whole vector $\boldsymbol{\theta}$ (denote all of the θ_i; $i = 1, 2, \ldots, k$ as the vector $\boldsymbol{\theta}$), a PEB approach models the θ_i's as random variables representing an unobserved process. In the engineering literature, this is a "signal-to-noise" problem, and it also arises in image analysis, where the goal is to uncover $\boldsymbol{\theta}$, called the hidden "image." Once $\boldsymbol{\theta}$ is uncovered, edges as well as other summaries can be computed. Consider the case where the set of θ_i's consists of a two-phase patchwork marked by discontinuities; e.g., $\theta_i = \alpha$ or $\theta_i = \beta$ for all i. That this is a reasonable model for the data is evidenced by Fig. 4, which shows a smoothed density estimate of the data. The data are bimodal, and appear to be a mixture of two bell-shaped distributions. The two-phase pattern contains change-points (or edges). A change-point is any point along

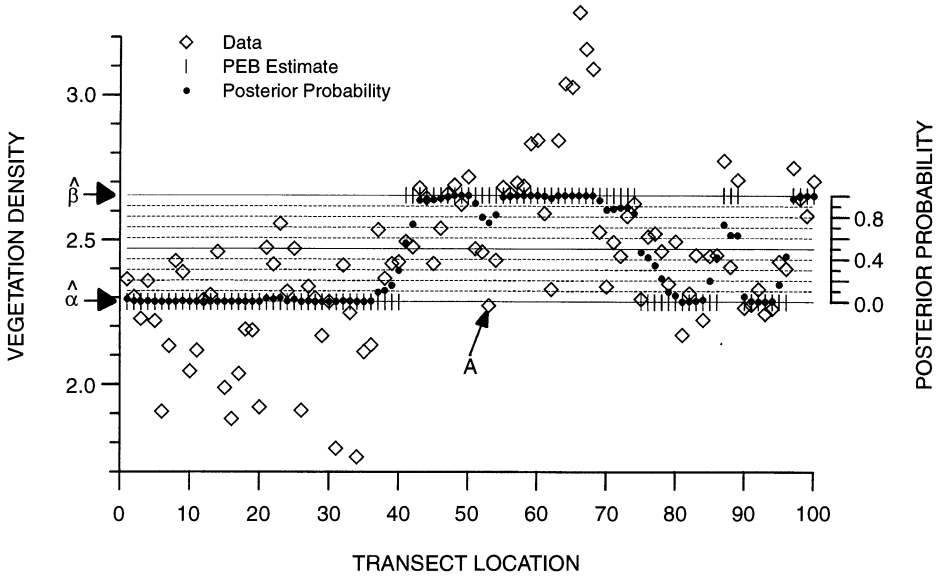

FIG. 3. Parametric empirical Bayes estimates for the first 100 plots of the line transect data. The data values ◇ are ln(% of nonwhite pixels + 1). The estimates $\hat{\alpha}$ and $\hat{\beta}$ are given on the left ordinate. The estimate for each θ_i is either $\hat{\alpha}$ or $\hat{\beta}$, and is given as a vertical bar. The posterior probability that $\hat{\theta}_i = \hat{\beta}$ is given by the solid circle read off the right ordinate.

the transect where the mean changes from α to β or vice versa.

Looking at Fig. 4, the most obvious way to proceed is to choose $\hat{\alpha}$ to be equal to the maximum at the first peak, and $\hat{\beta}$ to be the maximum at the second peak, and then let $\hat{\theta}_i = \hat{\alpha}$ if y_i is closer to $\hat{\alpha}$ or $\hat{\theta}_i = \hat{\beta}$ if y_i is closer to $\hat{\beta}$. However, this takes no account of the fact that there is some "smoothness" in the data. That is, occasionally a random variable with a mean of α may produce a value closer to β than α. However, if most of its neighbors had values closer to α, it seems reasonable to classify that random variable with a mean of α rather than β. In fact, with the proper prior distribution, this is what the PEB approach does. The estimates of α, β, and all θ_i for the first 100 data points in the transect are given in Fig. 3. Notice in particular

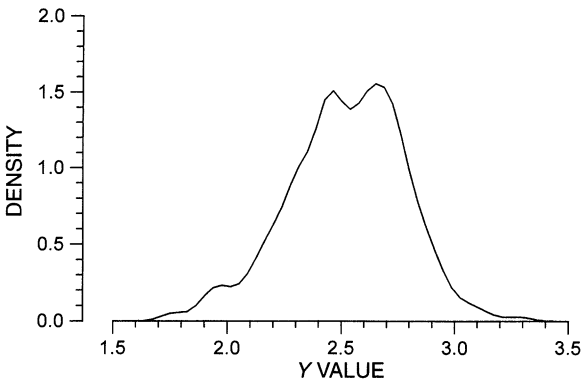

FIG. 4. Nonparametric density estimate for line transect data. The S-plus density function was used with a window width of 0.2 for smoothing.

that there is strong evidence for rather large patches. Thus, at the location indicated by A, the y value is below the estimate of $\hat{\alpha}$, yet because so many of its neighbors had values closer to $\hat{\beta}$, $\hat{\theta}_i$ at location A was estimated to be $\hat{\beta}$ rather than $\hat{\alpha}$. Along with the estimates, "credibility" statements about how likely θ_i is to be $\hat{\beta}$ or $\hat{\alpha}$ are also given. For example, again look at location A. The posterior probability that $\hat{\theta}_i$ is $\hat{\beta}$ is given by the solid circle read off the right ordinate, which is about 0.74. Thus, although $\hat{\theta}_i$ was estimated to be $\hat{\beta}$, we are only about 74% certain that was the right classification. Also note that much of the uncertainty occurs at the border between patches.

As before, it is possible to make estimates of functions of the parameters. For example, we might be interested in the "average patch size." For example, in a transect with 10 plots, if the first 5 had a mean of α, the next 3 had a mean of β, and the last 2 had a mean of α, then the average patch size would be $(5 + 3 + 2)/3$. Generalizing, average patch size might be defined as $\tau = n/(\text{number of patches}) = n/(\text{number of change-points} + 1)$. Calculating τ for 2000 Monte Carlo simulations from the posterior distribution yielded an average patch size of 14.71, with 95% credibility intervals ranging from 12.00 to 17.65.

STATISTICAL THEORY AND PHILOSOPHY

A parametric empirical Bayes analysis requires a hierarchical Bayesian setup using conditional probability. It is often easier to specify complex situations, which typically occur for ecological data, by using conditioning. The simplest hierarchical Bayesian setup

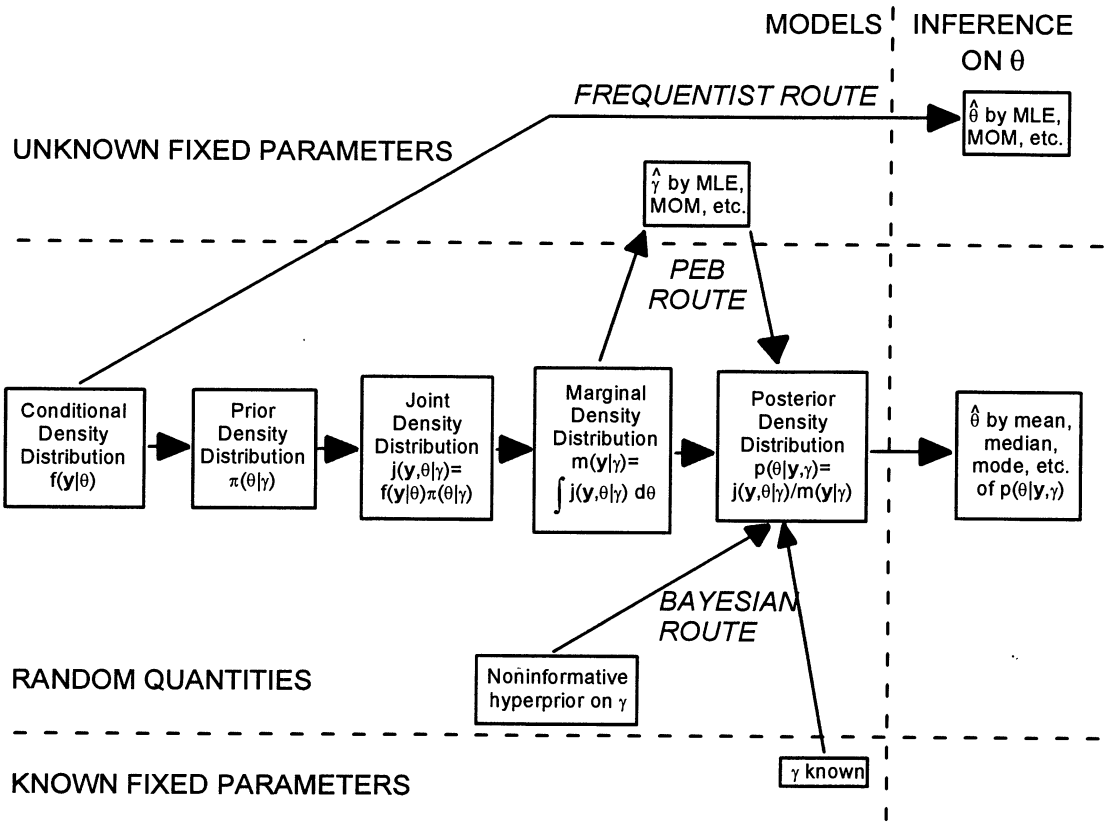

FIG. 5. Comparison of methods and philosophy employed by frequentist, Bayesian, and parametric empirical Bayesian.

contains two stochastic processes, one for the data, and one for the parameters. This is apparent in all three examples above. The data are observed, but not the parameters. Denote the probability distribution function of the data as $f(\mathbf{y} \mid \boldsymbol{\theta})$, with data $\mathbf{y}$ *conditional* on the "parameters" of interest $\boldsymbol{\theta}$. In the introduction, I asked whether it is possible to borrow information from other populations in estimating θ_i. The Bayesian does this by putting a distribution on the θ_i's, which ties them together. Hence, a Bayesian does not consider θ_i to be a fixed parameter, but assumes $\boldsymbol{\theta}$ to come from some prior distribution, call it $\pi(\boldsymbol{\theta} \mid \boldsymbol{\gamma})$, which itself depends on a vector of "hyperparameters" $\boldsymbol{\gamma}$. The notation $\pi(\boldsymbol{\theta} \mid \boldsymbol{\gamma})$ indicates that π is a probability distribution function (e.g., Normal, Poisson, etc.) of $\boldsymbol{\theta}$ with $\boldsymbol{\gamma}$ considered given or fixed. Then the joint density is

$$j(\mathbf{y}, \boldsymbol{\theta} \mid \boldsymbol{\gamma}) = f(\mathbf{y} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta} \mid \boldsymbol{\gamma}), \tag{2}$$

and from Bayes Theorem the posterior density of $\boldsymbol{\theta}$, given the data, is

$$p(\boldsymbol{\theta} \mid \mathbf{y}, \boldsymbol{\gamma}) = \frac{j(\mathbf{y}, \boldsymbol{\theta} \mid \boldsymbol{\gamma})}{\int j(\mathbf{y}, \boldsymbol{\theta} \mid \boldsymbol{\gamma})\, d\boldsymbol{\theta}}. \tag{3}$$

If the distribution of $\boldsymbol{\theta}$ is discrete, replace the integral in Eq. 3 with the appropriate sum. All inference proceeds from the posterior distribution. For example, the mean of the posterior is a Bayes estimate (under squared error loss), and credibility regions (Bayesian term for confidence intervals) can be taken directly from the posterior distribution; i.e., a lower 95% credibility endpoint l is, $\int_{-\infty}^{l} p(\boldsymbol{\theta} \mid \mathbf{y}, \boldsymbol{\gamma})d\boldsymbol{\theta} = 0.025$ and an upper 95% credibility endpoint u is $\int_{-\infty}^{u} p(\boldsymbol{\theta} \mid \mathbf{y}, \boldsymbol{\gamma})d\boldsymbol{\theta} = 0.975$.

Parametric empirical Bayes vs. other approaches

Fig. 5 is given as an aid in comparing PEB to the classical frequentist and Bayesian approaches. By "frequentist" I mean one who uses the usual sampling-based or model-based approaches of regression, ANOVA, etc., that we all learn in introductory statistics courses. The frequentist considers $\boldsymbol{\theta}$ to be a fixed, nonrandom quantity, and turns immediately from the conditional specification $f(\mathbf{y} \mid \boldsymbol{\theta})$ to estimation of $\boldsymbol{\theta}$ using maximum likelihood (MLE), method of moments (MOM), or other estimation procedures. For the Bayesian formulation, the question arises: What to do about hyperparameters $\boldsymbol{\gamma}$? The classical Bayesian approach is to allow "prior" or "personal" information into the problem, either by specifying the hyper-parameters $\boldsymbol{\gamma}$ of the prior distribution or putting a noninformative prior on them (a third level in a hierarchical model).

Now, notice that the integral,

$$\int j(\mathbf{y}, \boldsymbol{\theta} \mid \boldsymbol{\gamma})\, d\boldsymbol{\theta} = m(\mathbf{y} \mid \boldsymbol{\gamma}), \tag{4}$$

is the marginal density of $\mathbf{y}$, the data, and the only other parameters are $\boldsymbol{\gamma}$. The PEB approach is to use $m(\mathbf{y} \mid \boldsymbol{\gamma})$ to estimate $\boldsymbol{\gamma}$, rather than specifying it, and then plug the estimate of $\boldsymbol{\gamma}$ into the posterior distribution (Eq. 3). For example, maximum likelihood can often be used to estimate $\boldsymbol{\gamma}$ from Eq. 4. The main difficulty is that the integral or sum to obtain the marginal distribution (Eq. 4) may be difficult to obtain analytically. Numerical techniques such as the EM-algorithm (Dempster et al. 1977, Tanner 1993:38–57) have helped solve some of the problems of intractable marginal distributions, and was used, for example, in the grassland transect example.

The empirical Bayesian uses the Bayesian mathematical framework, but dips into frequentist philosophy to estimate all unknown parameters ($\boldsymbol{\theta}$ is not really a parameter because it is considered random). This has caused Lindley to say "there is no one less Bayesian than an empirical Bayesian." (See reference in Morris 1983*a*.) Yet PEB methods also seem to be greeted with skepticism by frequentists because of the Bayesian framework. One must draw one of two conclusions: either the method is truly bad, or it is truly a compromise between the two philosophies. It seems to me that when proper care is taken to ensure that the uncertainty in estimating the unknown parameters $\boldsymbol{\gamma}$ is properly accounted for in the posterior distribution, as Morris (1983*a, b*) does, the PEB approach combines the best of both philosophies. It has good long-run properties (e.g., the simulation experiment for the capture–recapture example), which the frequentist demands, yet has almost the power of the Bayesian method (the posterior distribution must be inflated) and the utility of the posterior distribution.

A detailed discussion of confidence intervals for PEB methods is beyond the scope of this paper. Basically, the posterior distribution needs to be "inflated" to account for the fact that $\boldsymbol{\gamma}$ is estimated, rather than known. This is especially true for small sample sizes. The interested reader is referred to Morris (1983*a, b*) and Laird and Louis (1987).

Theory applied to examples

In the examples above, $f(\mathbf{y} \mid \boldsymbol{\theta})$ was multivariate normal, assuming conditional independence. That is, given $\boldsymbol{\theta}$, $f(\mathbf{y} \mid \boldsymbol{\theta})$ is the product of independent normal distributions for each y_i. Then, in the first two examples, $\boldsymbol{\theta}$ itself was considered multivariate normal with mean $\mathbf{X}\boldsymbol{\beta}$, where $\mathbf{X}$ is a matrix with observed covariates and $\boldsymbol{\beta}$ is a parameter vector (the usual linear model notation for regression, ANOVA, etc.), and with variance-covariance matrix $\delta^2\mathbf{I}$, with $\mathbf{I}$ being the identity matrix. The harbor seal example relied on the estimate $\hat{\theta}_i$ having a normal distribution because it was a sum of 25 different sites. For the capture–recapture example, $\hat{\theta}_i$ is not normal, but $1/\hat{\theta}_i$ is more nearly normal. Inference was performed for $1/\theta_i$, and then inverted. In the transect example, $\boldsymbol{\theta}$ was given a Gibbs distribution, which is also a first-order Markov chain. In all cases, the main interest was inference on $\boldsymbol{\theta}$, and estimation of all "other" parameters proceeded from the marginal distribution (Eq. 4). More details are found in the Appendix.

Discussion and Conclusions

Hopefully, the wide diversity of these applications shows the generality and utility of parametric empirical Bayes methods. What defines parametric empirical Bayes is the focus on estimating the unknown $\boldsymbol{\theta}$ by putting a distribution on them, with estimation of all "other" unknown parameters from the marginal distribution (Eq. 4). There are relationships between PEB methods and repeated measures, random coefficients models, mixture models, and meta-analysis, which also use a hierarchical setup. These methods also often estimate parameters from the marginal distribution (Eq. 4), but usually the focus is on the hyperparameters, such as $\boldsymbol{\beta}$ in the linear model part of the prior distribution.

Other biological applications of PEB methods include population estimation for bowhead whales (Raftery et al. 1988) and estimating species richness (Mingoti and Meeden 1992). Applications in other subject areas include spatial models (Mollie and Richardson 1991), count data as event rates (Gaver and O'Muircheartaigh 1987), and finite population estimation (Nandram and Sedransk 1993).

Computer programs remain a stumbling block. General programs for hierarchical models have been reviewed by Kreft et al. (1994). The first two examples analyzed here use the S-plus function GRIMM, by Everson and Morris, obtained by anonymous ftp (see *Acknowledgments*). The first example uses GRIMM without modification. For the capture–recapture example, because of the inverting, GRIMM was used as part of a larger capture–recapture S-plus function. For the transect example, FORTRAN programs were written.

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APPENDIX
MATHEMATICAL DETAILS

Harbor seal example

Assume that

$$\hat{\theta}_i \mid \theta_i \sim N(\theta_i, \sigma_i^2) = f(\hat{\theta}_i \mid \theta_i, \sigma_i^2) \qquad (A.1)$$

where $\hat{\theta}_i$ is the bootstrap estimate in the harbor seal example. Next, suppose that trend among years is modeled linearly. The average number of seals hauled out during the molting period for each year will not fall exactly on this trend line, so the deviations from the trend line are modeled as

$$\theta_i \sim N(\beta_0 + \beta_1 i, \delta^2) = \pi(\theta_i \mid \beta_0, \beta_1, \delta^2) \qquad (A.2)$$

where θ_i is now considered a random variable for the average number of seals hauled out in year i, and $N(\beta_0 + \beta_1 i, \delta^2)$ is a normal distribution with a trending mean $\beta_0 + \beta_1 i$ and variance δ^2. As stated in the introduction, the PEB approach requires estimation of the prior parameters $\gamma = (\beta_0, \beta_1, \delta^2)'$ from the marginal distribution of $\hat{\theta}_i$. Jointly, $(\hat{\theta}_i, \theta_i)$ are distributed as $f(\hat{\theta}_i \mid \theta_i, \sigma_i^2)\pi(\theta_i \mid \beta_0, \beta_1, \delta^2)$ (Eq. 2). The marginal distribution $m(\hat{\theta}_i \mid \beta_0, \beta_1, \delta^2)$, unconditional on θ_i, has a distribution (Morris 1983*a*),

$$\hat{\theta}_i \sim N(\beta_0 + \beta_1 i, \sigma_i^2 + \delta^2). \qquad (A.3)$$

The values $\hat{\theta}_i$ and σ_i^2 are known, and Morris (1983*a*) shows how to estimate β_0, β_1, δ^2. The posterior distribution of $\theta_i \mid \hat{\theta}_i$ is

$$p(\theta_i \mid \hat{\theta}_i, \sigma_i^2, \delta^2, \beta_0, \beta_1) \sim N[A_i, \sigma_i^2(1 - B_i)], \qquad (A.4)$$

where $A_i = (1 - B_i)\hat{\theta}_i + B_i\mu_i$, $B_i = \sigma_i^2/(\sigma_i^2 + \delta^2)$ and $\mu_i = \beta_0 + \beta_1 i$. Although it would be possible to make inference by maximum likelihood estimation for β_0, β_1, and δ^2 from Eq. A.3, and then by "plugging" estimates of σ_i^2, β_0, β_1, and δ^2 into the posterior distribution (Eq. A.4), Morris (1983*a*) uses some small sample theory to ensure that "overshrinking" does not occur and that proper inflation of the posterior occurs.

Capture–recapture example

For single capture–recapture experiments, Jensen (1989) develops confidence intervals for $1/\hat{\theta}_i$ based on a normal distribution, and then inverts them to get confidence intervals for θ_i. Therefore, the PEB estimates were also obtained for $1/\theta_i$ in order to use the normal theory, and then inverted back to the original scale. The PEB estimates were obtained by shrinking the estimates $1/\hat{\theta}_i$ toward the regression line using the same theory as given above for the harbor seal example, using the estimated variance for $1/\hat{\theta}_i$ as given by Jensen (1989). Some care is necessary to ensure that the estimates remain unbiased. A Taylor series expansion shows that if $E(X) = \mu$ and $\mathrm{var}(X) = \sigma^2$, then $E(1/X) \doteq 1/\mu + \sigma^2/\mu^3$. Thus, the Chapman estimates were inverted by $\widehat{1/\theta_i} = 1/\hat{\theta}_i - \hat{\sigma}_c^2/\hat{\theta}_i^3$, where $\hat{\sigma}_c^2$ is given by Seber (1970),

$$\sigma_c^2 = \frac{(\hat{\theta}_i + 1)(m_i - r_i)(c_i - r_i)}{(r_i + 1)(r_i + 2)}.$$

After shrinking the $\widehat{1/\theta_i}$ toward the regression line, call these estimates $\widetilde{1/\theta_i}$, the bias correction is again used to convert back to the original scale: $\tilde{\theta}_i = 1/(\widetilde{1/\theta_i}) - [\mathrm{SE}(\widetilde{1/\theta_i})]^2/(\widetilde{1/\theta_i})^3$, where SE is the standard error of the PEB estimates. The confidence intervals were developed for $\widetilde{1/\theta_i}$ based on normal theory, and inverted with the same bias correction as for the PEB estimate. Because the regression line is a quadratic, it also needs to be inverted.

Transect example

Assume the following model. All random variables y_i are conditionally independent and identically distributed from a Gaussian distribution

$$f(\mathbf{y} \mid \boldsymbol{\theta}, \sigma^2, \alpha, \beta) = \prod_{i=1}^{n} f(y_i \mid \theta_i, \sigma^2, \alpha, \beta) \qquad (A.5)$$

where $f(y_i \mid \theta_i, \sigma^2, \alpha, \beta)$ in Eq. A.5 is a normal distribution with variance σ^2, mean θ_i that can only take values α or β. For the prior, consider

$$\pi(\boldsymbol{\theta}; \lambda) = \exp[-\lambda c(\boldsymbol{\theta})]/[2(\exp(-\lambda) + 1)^{n-1}] \qquad (A.6)$$

where $c(\boldsymbol{\theta})$ is the number of change-points: $c(\boldsymbol{\theta}) = \sum_{i=1}^{n-1} I(\theta_i \neq \theta_{i+1})$, I being the indicator function, equal to 1 when $\theta_i \neq \theta_{i+1}$ is true, and equal to 0 when $\theta_i \neq \theta_{i+1}$ is false. The prior distribution (Eq. A.6) is both a first-order Markov chain and a first-order Gibbs chain (Qian and Titterington 1990). It is a slight generalization of priors studied by Greig et al. (1989), and Qian and Titterington (1990, 1991), where they use $\alpha = 0$ and $\beta = 1$. The parameter λ is a "smoothing" parameter that, as λ gets larger, puts relatively more weight on fewer change-points. The marginal density of the data $\mathbf{y}$ is

$$h(\mathbf{y} \mid \sigma^2, \alpha, \beta, \lambda) = \sum_{\theta_1 \in A} \sum_{\theta_2 \in A} \cdots \sum_{\theta_n \in A} f(\mathbf{y} \mid \boldsymbol{\theta}, \sigma^2, \alpha, \beta)\pi(\boldsymbol{\theta} \mid \lambda), \qquad (A.7)$$

where $A = \{\alpha, \beta\}$ and the posterior density of the image $\boldsymbol{\theta}$, given the data, is,

$$p(\boldsymbol{\theta} \mid \mathbf{y}, \sigma^2, \alpha, \beta, \lambda) = f(\mathbf{y} \mid \boldsymbol{\theta}, \sigma^2, \alpha, \beta)\pi(\boldsymbol{\theta} \mid \lambda)/h(\mathbf{y} \mid \sigma^2, \alpha, \beta, \lambda). \qquad (A.8)$$

The EM-algorithm for obtaining estimates of α, β, σ^2, and λ from the marginal distribution (Eq. A.7) is developed in Qian and Titterington (1990). The Bayes rule chosen was the maximum of the posterior distribution (Eq. A.8), which was obtained using simulated annealing (Cressie 1991:512). In order to simulate from the posterior distribution, the Gibbs sampler was used (Cressie 1991:571, Tanner 1993:102–146).